SALTDRAG · TRACEABILITY MATRIX

SaltDrag traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
Source	docs/03b-traceability.md @ c7f0938		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

SaltDrag traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 15 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Work from the bund	SDG-CAL-001	Analysis (layout), TRL 3; observed field trial on a partner pan, TRL 5	Met by design
UR-002	PR-002	Crank effort	SDG-CAL-001, SDG-BLD-001, SDG-STM-001	Analysis (CAL-001 [B3]), TRL 3; spring scale on the handle, TRL 4	Met on paper
UR-003, UR-009	PR-003	Pull capacity and margin	SDG-CAL-001, SDG-BLD-001, SDG-STM-001	Analysis (CAL-001 [B4], [B5], Section C), TRL 3; proof load with CalRig, TRL 4	Met on paper
UR-004	PR-004	Protect the pan floor	SDG-CAL-001, SDG-BLD-001, SDG-STM-001	Analysis (CAL-001 [D1], [D2]), TRL 3; inspection after 20 passes, TRL 5	Met by design
UR-003	PR-005	Gathering rate	SDG-CAL-001, SDG-OPT-001	Side-by-side timed harvest trial, TRL 5	Not met (not verified; no baseline)
UR-005	PR-006	Part mass	SDG-CAL-001, SDG-BLD-001	Analysis (model masses, CAL-001 [E1]), TRL 3; weigh parts, TRL 4	Met on paper
UR-006	PR-007	Corrosion resistance	SDG-CAL-001, SDG-STM-001	Analysis (materials, SDG-STM-001 S18 to S20), TRL 3; season trial with inspection log, TRL 5	Met by design (316 stainless in brine to confirm, S19)
UR-007	PR-008	Quick set-up	SDG-CAL-001, SDG-BLD-001	Analysis (CAL-001 [D10]), TRL 3; timed trials, TRL 4	Met on paper, small margin
UR-008	PR-009	Value engineering	SDG-CAL-001	Analysis (costed BOM, CAL-001 [E2]), TRL 3	Met on paper (USD 2,897)
UR-009	PR-010	Rope safety	SDG-CAL-001, SDG-BLD-001, SDG-STM-001	Analysis (CAL-001 [C8]), TRL 3; rope certificate and proof load, TRL 4	Met on paper
UR-009	PR-011	Holding	SDG-CAL-001, SDG-BLD-001, SDG-STM-001	Analysis (CAL-001 [C5b], [C7], [C10b]), TRL 3; proof load and anchor pull test on site, TRL 4	Met on paper; soil to confirm
UR-012	PR-012	Pan coverage	SDG-CAL-001, SDG-BLD-001, SDG-STM-001	Analysis (CAL-001 [B8]), TRL 3; wind on and measure, TRL 4	Met on paper
UR-010	PR-013	Hand power only	SDG-CAL-001	Inspection of the design (CAL-001 Section G), TRL 3	Met by design
UR-013	PR-014	Nothing left in the sanctuary	SDG-CAL-001	Inspection of the design (CAL-001 Section G; SDG-BLD-001), TRL 3; demonstration at the trial pan, TRL 5	Met by design

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-011	PR-015	Local stock	SDG-CAL-001	Inspection of the BOM (CAL-001 Section G), TRL 3; build, TRL 4	Met by design
UR-014	PR-016	Crew size	SDG-CAL-001	Analysis (CAL-001 [D9], Section G), TRL 3; trials, TRL 5	Met by design
UR-015	PR-017	Daily care	SDG-CAL-001	Inspection of the BOM and build plan (CAL-001 Section G), TRL 3; season trial, TRL 5	Met by design
UR-002, UR-009	PR-018	Cranking power	SDG-CAL-001, SDG-STM-001	Analysis (CAL-001 [B7b]; SDG-STM-001 S15), TRL 3; heart-rate and timing trial with the partner, TRL 5	Met on paper (71 W)

Gaps

None.